Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an Explainable (XAI) Intelligent Quality Inspection System based on Multi-modal Sensing for Metal Forming and Welding Processes**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Machinery and materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Casting/Welding	*Other Casting/Welding Technology*	*Shipbuilding/Marine systems*	*Ship Materials/Structural Technologies*
Concept of Proposed Technology	**Core Concept** The implementation of multimodal sensing fusion in shipbuilding provides a decisive competitive advantage by shifting from reactive to proactive quality assurance. By detecting internal micro-cracks and surface flaws during the assembly of hull blocks, shipyards can significantly reduce the astronomical costs associated with post-delivery repairs and dry-dock rework, which often run into millions of dollars. The integration of **large language model (LLM)**-based XAI further streamlines operations by reducing the reliance on a shrinking pool of highly specialized **NDT (Non-Destructive Testing)** experts; the system provides immediate, standards-compliant feedback that allows floor technicians to perform high quality welding. Ultimately, this technology minimizes project delays, ensures compliance with strict maritime classification societies, and enhances the lifetime safety of high-value vessels like **liquified natural gas (LNG)** carriers and deep-sea research ships.			

	Key Functions and Potential Applications The proposed system architecture consists of a high-frequency **Acoustic Emission (AE)** sensor array (operating in the 100 kHz to 1 MHz range) paired with 4K high-speed computer vision cameras mounted on the welding/press carriage. The data backbone utilizes a sensor fusion algorithm that time-syncs acoustic "hits" with visual frame timestamps to correlate internal stress events with surface morphology. The diagnostic engine is powered by a **Deep Neural Network (DNN)** for defect classification, which feeds into an **Explainable AI (XAI)** layer. This layer utilizes **Gradient-weighted Class Activation Mapping (Grad-CAM)** to generate spatial heatmaps of defects. Finally, a specialized LLM, fine-tuned on shipbuilding technical codes (e.g., AWS D1.1 or ISO 5817), parses the fused data to output natural language root-cause analyses and recommended corrective actions via a real-time dashboard.
Rationale for Joint R&D Project	Rationale for the Joint R&D project can be listed as follows. **1. Overcoming Single-Sensor Limitations and Enhancing Process Intelligence** • Multimodal Platform Necessity: Traditional vision inspection is incapable of detecting internal metallic cracks, while standalone AE sensors struggle with precise localization of surface defects. Developing a multimodal platform that integrates both is essential to eliminate these blind spots. • Transition to Autonomous Manufacturing: To move beyond simple detection toward "Autonomous Manufacturing"—where process anomalies are diagnosed and rectified independently—it is critical to adopt advanced global technologies such as XAI and LLM. **2. Addressing Data Imbalance and Accelerating Commercialization** • Data Augmentation via GANs: Real-world manufacturing environments suffer from a scarcity of "defect" data. By leveraging **Generative Adversarial Network (GAN)** technology from Purdue University, we can synthesize and augment virtual defect data to ensure the high reliability and accuracy of our AI models. • From Lab to Market: This collaboration aims to merge Purdue University's foundational algorithms with the hardware manufacturing expertise of Korean Small and medium enterprises (SMEs). The goal is to rapidly transition laboratory-scale technology into a market-ready product for immediate commercial application. **3. Necessity of global Collaboration** • Purdue University, a leading overseas research institution, is a manufacturing-focused engineering university located in Indiana, a region surrounded by numerous manufacturing companies, and serves as an industrial hub in the United States. The university has already been collecting meaningful equipment signal data and IoT sensing data from major manufacturing corporations in the surrounding area and utilizing them for AI-based research. • Purdue maintains strong collaborative relationships with medium- to large-scale enterprises equipped with factories, facilities, and production lines;

	• it operates specialized research centers capable of conducting testbed-based research on manufacturing and physical AI; • it is strategically located in a major transportation hub in the U.S., making it the only academic institution where Korean companies can directly identify technological trends and market demands in the local environment. Therefore, global collaboration is essential.
Key Words	**1. Welding inspection system** **2. Multi-model inspection** **3. Explainable AI** **4. Defect detection** **5. Synthetic data**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

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This project aims to develop a real-time defect detection and explanation system for steel sheet forming (press) and welding processes by integrating AE sensing with high-resolution vision data. The proposed multimodal sensor fusion approach enables the simultaneous identification of internal micro-cracks and external surface defects, overcoming the limitations of single-modality inspection methods. To enhance trust and usability in industrial environments, the project incorporates XAI that not only detects defects but also visualizes their locations using heatmaps and automatically generates natural-language explanations of defect causes through an LLM–based agent, grounded in established technical standards. This capability supports informed decision-making and accelerates deployment in smart manufacturing systems.

R&D Target Specifications

Performance Indicator	Unit	Target	Domestic Best	Global Best	Remark
1.Defect Detection Accuracy	%	≥ 95	-	-	AE/Vision
2. Real-time Latency	ms	< 200	-	-	Per item (Edge)
3. Data Augmentation	cases	≥ 5,000	-	-	GAN
4. Analysis Reliability	%	≥ 90	-	-	XAI/LLM

R&D Plan

Multi-modal Fusion:

This work develops a heterogeneous data fusion algorithm that integrates high-frequency AE signals, which are sensitive to internal defects, with vision-based image data used for detecting surface defects. By jointly analyzing these complementary sensing modalities, the proposed approach improves defect detection robustness and accuracy compared to single-sensor methods.

Generative AI:

To address the challenge of limited defect data, the project employs GANs to synthesize high-quality artificial defect images. These synthetic datasets are used to augment training data, enhancing model generalization and performance under data-scarce industrial conditions.

Edge AI Optimization:

For deployment in real manufacturing environments, the project incorporates edge AI optimization techniques, including model quantization and TensorRT-based acceleration, to achieve lightweight, low-latency inference on industrial edge devices while maintaining detection accuracy.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Purdue University**: Applied Optics lab from the School of Mechanical Engineering will contribute to this project on the sensor network design and data analysis part using various machine learning algorithm and expanding to XAI domain. The team has extensive experience in merging engineering hardware system with machine learning algorithms such as linear discriminant analysis (LDA), artificial neural network (ANN), support vector machine (SVM) on elastic light scatter pattern analysis, mushroom classifications, and electronic nose data for classifications etc [1–7]. In addition, with the emphasis in system, measurement, and control area within Mechanical engineering, PI will work closely with Daemyung corporation in sensor design and non-destructive testing system design through various capstone design projects to provide foundational concept generation and down selection process, risk-reward analysis, and potential connection to Purdue's Mitch Daniel Business school for marketing analysis for the future projects for commercialization.
Expected R&R of Korean SME Partner	• **Korean SME partner**: "Hardware System Construction, Field Validation, and Process Optimization" **HW System Design & Integration** Inspection Equipment Fabrication: Design and manufacturing of optimized inspection hardware (Housing/Jig) equipped with high-sensitivity AE sensors, specialized lighting (Polarized/HDR), and high-resolution cameras. System Integration: Development of middleware and interfaces (OPC-UA/Modbus) to link AI Edge Boxes with existing PLC/MES production systems. **Validation & Process Optimization** Test-bed Operation & Data Collection: Provision of actual production lines as test-beds for raw data collection and field validation of AI models.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Achievements: Establishing an Integrated Platform and Resolving Data Scarcity** • Successfully secured an integrated inspection platform capable of detecting both internal and external defects with over 95% accuracy. • Effectively resolved the data scarcity issue (the lack of real-world defect data) by utilizing synthetic data generation techniques. **Economic and Industrial Impact: Enhanced Efficiency and Revenue Growth** • Achieved a five-fold (5x) increase in inspection speed compared to manual or single-sensor methods. • Projected to reduce annual quality-related costs by 30%. • Anticipated to drive a revenue increase of 12% or more for the participating Small and Medium Enterprise (SME) partner. **Workplace Improvement: Safety and Strategic Workforce Reallocation** • Enhanced worker safety by automating inspections in hazardous press-forming and welding environments. • Enabled the reallocation of human resources from repetitive manual inspection to high-value-added strategic tasks.

6. References

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